

KNOX GRAMMAR SCHOOL



TRIAL H.S.C. EXAMINATION

1994

MATHEMATICS

2 UNIT

YEAR 12

*Time allowed: Three hours
(plus five minutes reading time)*

INSTRUCTIONS

- ALL* questions should be attempted.
- ALL* questions are of equal value.
- ALL* necessary working should be shown in every question.
- Full marks may not be awarded if work is careless or badly arranged.
- Approved calculators may be used.
- Each question should be started on a new page.

The papers are to be handed in the following manner:

Part A	Questions 1-2
Part B	Questions 3-4
Part C	Questions 5-6
Part D	Questions 7-8
Part E	Questions 9-10

Name: _____ Class: _____

PART A

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QUESTION 1 12 marks

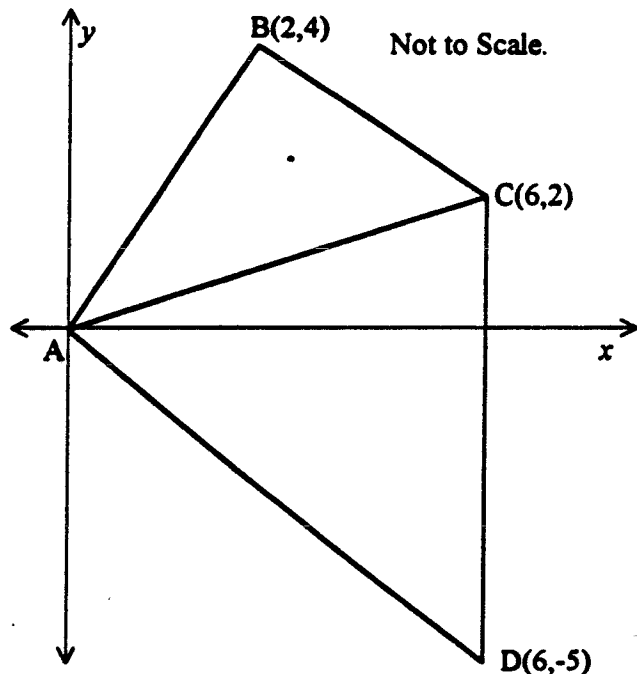
- (a) Given $R = \frac{\pi(D+3K)}{K-2D}$ where $D = 0.0034$ and $K = 0.087$, find the value of R correct to three significant figures.
- (b) Write down the gradient of the line $2x + 3y = 5$.
- (c) Solve $16^2 = 2^{2t-3}$ for t .
- (d) Solve the pair of simultaneous equations:
$$\begin{aligned} 2x + 3y &= 9 \\ x - 2y &= 1. \end{aligned}$$
- (e) Graph on a number line the solution set of $-12 \leq \frac{3}{2}x < 3$.
- (f) Solve the equation $5x^2 - 4x - 4 = 0$ giving each solution correct to two decimal places.

QUESTION 2 12 marks

Start a new page

In the diagram A , B , C and D have coordinates $(0,0)$, $(2,4)$, $(6,2)$ and $(6,-5)$ respectively.

- (a) Show $\triangle ABC$ is a right-angled isosceles triangle.
- (b) Find the gradient of AC .
- (c) Find the equation of AC .
- (d) Find the angle which the line AC makes with the positive x -axis.
- (e) Find the length of AC .
- (f) Find the perpendicular distance of the point D from the line AC .
- (g) Find the area of the quadrilateral $ABCD$.



PART B

QUESTION 3 12 marks

Start a new page

- (a) Differentiate:
- (i) $3 \tan(4x - \frac{\pi}{4})$
 - (ii) $x^2 e^{-2x}$
 - (iii) $\frac{\log_e 3x}{x}$
- (b) Evaluate $\int_0^1 5x + \sin(5x) dx$ to 2 decimal places.
- (c) Evaluate $\int_1^e \frac{2}{x} + \frac{x}{2} dx$ leaving your answer in exact form.

QUESTION 4 12 marks

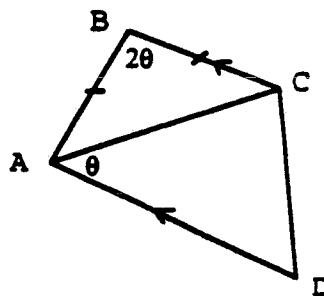
Start a new page

- (a) The vertex of a parabola is at $V(2, -1)$ and the focus is at $S(2, 3)$.
- (i) Sketch the parabola showing the directrix, vertex and focus.
 - (ii) Write down the focal length and the equation of the directrix.
 - (iii) Find the equation of the parabola.

- (b) In the diagram $ABCD$ is a quadrilateral with $AB = BC$,

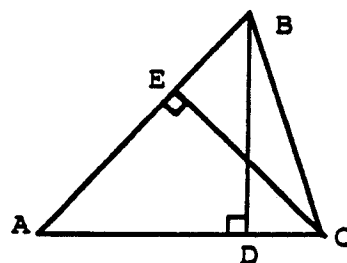
$BC \parallel AD$, $\angle ABC = 2\theta$ and $\angle CAD = \theta$.

Copy this diagram on your answer sheet and find the value of θ giving reasons.



- (c) In the diagram $BD \perp AC$ and $CE \perp AB$.

- (i) Copy this diagram on your answer sheet and prove that $\triangle ECA$ is similar to $\triangle DBA$.
- (ii) If $AB = 10\text{cm}$, $BD = 7\text{cm}$ and $AC = 6\text{cm}$, find the length of CE .



PART C

QUESTION 5 12 marks

Start a new page

- (a) Consider the curve $y = x + \sin(2x)$ for $0 \leq x \leq \pi$.
- (i) Find the two turning points for $0 \leq x \leq \pi$.
 - (ii) Determine the nature of the turning points.
 - (iii) Show there is one point of inflection for x between 0 and π .
- (b) Sketch the curve of $y = e^{-2x}$ and shade the region bounded by this curve the x -axis, the y -axis and the line $x = 2$.

Find the volume generated when this area is rotated about the x -axis (leaving your answer in terms of π).

QUESTION 6 12 marks

Start a new page

- (a) The speed of the cyclists in the Commonwealth Games was recorded every ten minutes. A table was drawn up of the time, in minutes, and the corresponding speeds S , in km/hr.

Time (min)	0	10	20	30	40	50	60
Speed (km/h)	0	52	49	50	52	53	55

Use the Trapezoidal Rule to find the approximate value of $\int_0^1 S dt$ (where t is in hours).

- (b) Find the values of m for which the equation $2x^2 - mx + 18 = 0$ has real roots.
- (c) If α and β are the roots of the equation $2x^2 - 3x - 5 = 0$, find the value of
- (i) $\alpha + \beta$
 - (ii) $\alpha\beta$
 - (iii) $(\alpha + 1)(\beta + 1)$
- (d) David borrows \$ 20 000 at 18% p.a. reducible interest, and pays it off in equal monthly instalments over 5 years. Calculate the size of each instalment to the nearest cent.

PART D

QUESTION 7 12 marks

Start a new page

- (a) Explain why the series $\frac{9}{8}, \frac{3}{4}, \frac{1}{2}, \dots$ has a limiting sum. Find this limiting sum.
- (b) Given the sequence $U_n = 2n + 3$.
- (i) Find U_5, U_6, U_7 and U_{20} .
- (ii) Is this sequence a G.P. or an A.P.? Give reasons.
- (iii) Find $\sum_{n=5}^{20} (2n + 3)$.
- (c) Find a number 'n' which when added to each of 2, 5 and 9 will give a set of three numbers in geometric progression.

QUESTION 8 12 marks

Start a new page

- (a) A particle initially at rest at the origin moves in a straight line with velocity v metres per second, such that $v = 3t(4 - t)$, where t is the time elapsed in seconds. Find
- (i) the acceleration of the particle at the end of 1 second,
- (ii) an expression for the displacement x of the particle in terms of t ,
- (iii) the particle's displacement when it is next at rest,
- (iv) the velocity of the particle when it returns to the origin,
- (v) the time taken for the particle to reach its greatest velocity,
- (vi) the distance travelled by the particle in the first 5 seconds.
- (b) The point $P(x, y)$ moves such that its distance from the origin is twice its distance from the point $A(3, 3)$. Represent this information on a number plane and show that the locus of P is given by $(x - 4)^2 + (y - 4)^2 = 8$.

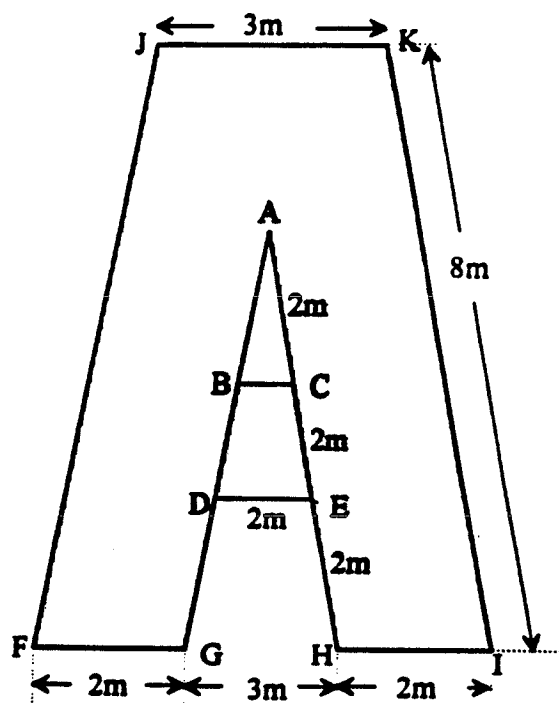
PART E

QUESTION 9

12 marks

Start a new page

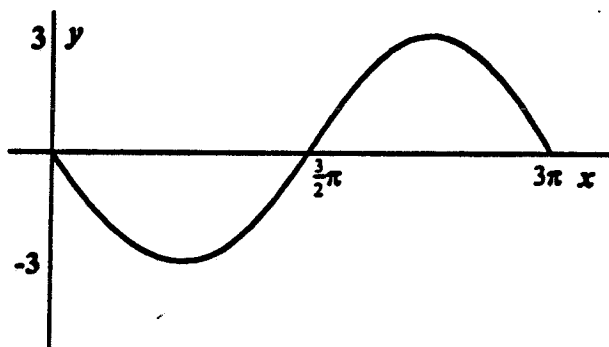
- (a) The letter **A** is to be painted on a billboard. The drawing opposite, not to scale, gives the dimensions of the letter : $JF = IK = 8\text{m}$, $JK = 3\text{m}$, and $AB, AC, BD, CE, DG, EH, FG, DE, HI$ all equal 2m .



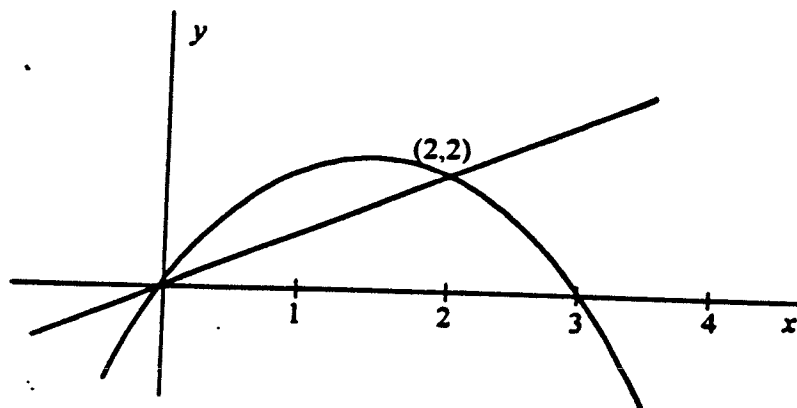
- (i) Find the size of $\angle DAE$, to the nearest minute, using the cosine rule.
- (ii) Find the length of BC .
- (iii) Show the vertical height between BC and DE is $\frac{\sqrt{15}}{2}$ m.
- (iv) Show the vertical height of the letter **A** is $2\sqrt{15}$ m.
- (v) Find the area of the letter **A** as a surd.

- (b) The diagram below represents a possible sine or cosine curve.

- (i) Give the amplitude.
- (ii) Give the period.
- (iii) Write down a possible equation for the curve.

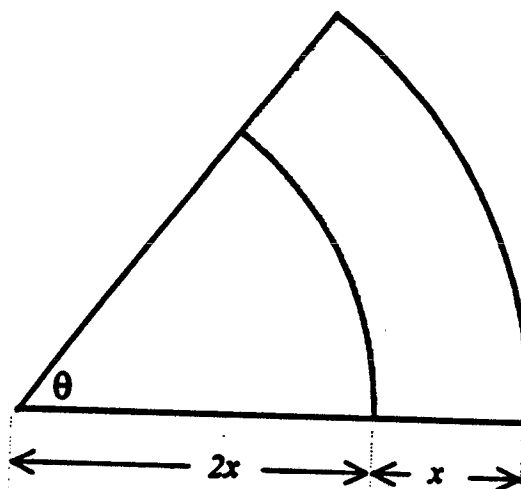


- (a) The graphs, below, are represented by $y = x(3 - x)$ and $y = x$.



- (i) Find the area between the curve $y = x(3 - x)$ and the x -axis.
 - (ii) Find the area between the curve $y = x(3 - x)$ and the line $y = x$.
 - (iii) What percentage of the area in part (i) is the area in part (ii)?
- (b) (i) Write down the formulae for the length l of an arc of a circle in terms of r and θ .
- (ii) Write down the formulae for the area A of a sector of a circle in terms of r and θ

- (c) The shaded area in the diagram represents a portion of an annulus.



- (i) Given the perimeter of the shaded region is 20 units, show the area of the shaded region is given by the formulae $A = 10x - x^2$.
- (ii) Find the maximum area of this region.

End of Paper.

Q1.2). $R = 10.357 = 10.4$ (3 s.f.).

b) $m = -\frac{2}{3}$

c) $2t - 3 = 8 \quad \therefore t = \frac{11}{2} \approx t = 5\frac{1}{2}$

d) $\begin{matrix} 2x + 3y = 9 \\ 2x - 4y = 2 \\ \hline 7y = 7 \end{matrix} \quad \therefore y = 1 \text{ and } x = 3.$

e) $-12 \leq \frac{3}{2}x < 3 \quad \therefore -8 \leq x < 2.$



f) $x = \frac{-4}{10} \pm \frac{\sqrt{16+80}}{10} = \frac{4}{10} \pm \frac{\sqrt{96}}{10}.$

$x = 0.4 + 0.97979 \quad \text{or } x = 0.4 - 0.97979$

$x = 1.38$ (2dp) $\text{or } x = -0.58$ (2dp)

Q2. a) $m_{AB} = \frac{4-0}{2-0} = 2 \quad m_{BC} = \frac{2-4}{6-2} = -\frac{1}{2}.$

$m_{AB} \cdot m_{BC} = 2 \cdot (-\frac{1}{2}) = -1 \quad \therefore AB \perp BC.$

$AB = \sqrt{2^2+4^2} = \sqrt{20}$
 $BC = \sqrt{(2-6)^2+(4-2)^2} = \sqrt{20}.$

$\therefore \Delta ABC$ is a right-angled isosceles Δ .

b) $m_{AC} = \frac{2-0}{6-0} = \frac{1}{3}.$

c) $y-0 = \frac{1}{3}(x-0) \quad \therefore y = \frac{1}{3}x \text{ or } x-3y=0.$

d) $\tan \theta = \frac{1}{3} \quad \therefore \theta = 18^\circ 26' \approx \theta = 18.435^\circ.$

e) $AC = \sqrt{6^2+2^2} = \sqrt{40} \approx 2\sqrt{10}$

d) $d = \left| \frac{1(6) - 3(-5)}{\sqrt{1^2+(-3)^2}} \right| = \frac{21}{\sqrt{10}}$

g)
 $A = 54 - (15+4+4) = 31 \text{ units}^2$
 $A = \frac{1}{2} 120 \times \sqrt{20} + \frac{1}{2} 2\sqrt{10} \times \frac{21}{\sqrt{10}}.$
 $= 10 + 21 = 31 \text{ units}^2.$

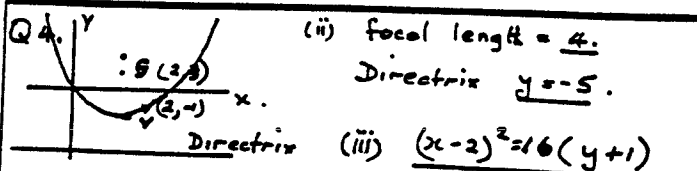
Q3. a) (i) $\frac{d}{dx} (3 \tan 4x - \frac{\pi}{2}) = 12 \sec^2(4x - \frac{\pi}{2}).$

(ii) $\frac{d}{dx} (x^2 e^{-2x}) = (2x)(e^{-2x}) + (-2e^{-2x})(x^2)$
 $= (2x - 2x^2) e^{-2x}.$

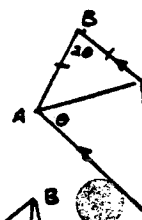
(iii) $\frac{d}{dx} \left(\frac{\log 3x}{x} \right) = \frac{x \left(\frac{3}{5x} \right) - (1) \log 3x}{x^2}$
 $= \frac{1 - \log 3x}{x^2}.$

b) $\int_0^1 (5x + \sin 5x) dx = \frac{5}{2} x^2 - \frac{1}{5} \cos 5x \Big|_0^1$
 $= \left[\frac{5}{2} - \frac{1}{5} \cos 5 \right] - \left[0 - \frac{1}{5} \cos 0 \right] = \frac{5}{2} - 0.0567 + \frac{1}{5}$
 $= 2.64$ (2dp).

c) $\int_1^e \left(\frac{2}{x} + \frac{x}{2} \right) dx = 2 \log x + \frac{1}{4} x^2 \Big|_1^e$
 $= \left[2 \log e + \frac{e^2}{4} \right] - \left[2 \log 1 + \frac{1}{4} \right] = 2 + \frac{e^2}{4} - \frac{1}{4}$
 $= \frac{1}{4} (7 + e^2) \approx 1\frac{3}{4} + \frac{1}{4} e^2.$



b) $\hat{BCA} = \theta$ (alternate angles of || lines)
 $\hat{BAC} = \theta$ base angles of isosc Δ .
 $2\theta + \theta + \theta = 4\theta = 180$ Sum angles of Δ
 $\therefore \theta = 45^\circ$

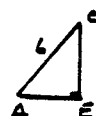
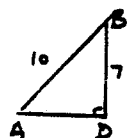


c) in ΔBDA and ΔCEA .

$\hat{BDA} = \hat{CEA}$ Both right angles.

$\hat{DAB} = \hat{ECA}$ common angle.

$\therefore \Delta BDA \parallel \Delta CEA$ A.A. $\approx$ (2 angles).



$\frac{CE}{6} = \frac{7}{10}$

$\therefore CE = \frac{42}{10} \approx 4.2 \text{ cm}.$

Q5. $y = x + \sin(2x) \quad \frac{dy}{dx} = 1 + 2\cos 2x \quad \frac{d^2y}{dx^2} = -4\sin 2x$

i) SP. $\frac{dy}{dx} = 0 \quad \therefore \cos 2x = -\frac{1}{2}, 2x = \frac{2\pi}{3} \text{ or } 2x = \frac{4\pi}{3}.$

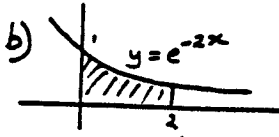
$\therefore$ SP. $x = \frac{\pi}{3}, y = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$ and $x = \frac{2\pi}{3}, y = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$

ii) at $x = \frac{\pi}{3}, \frac{d^2y}{dx^2} = -4\sin \frac{2\pi}{3} = -2\sqrt{3} \therefore$ C.C. down $\therefore$ M.F.F.

at $x = \frac{2\pi}{3}, \frac{d^2y}{dx^2} = -4\sin \frac{4\pi}{3} = 2\sqrt{3} \therefore$ C.C. Up $\therefore$ Min T.F.

iii) $\frac{d^2y}{dx^2} = 0 \quad \therefore \sin 2x = 0 \quad \therefore 2x = 0 \text{ or } 2x = \pi.$
 $\therefore x = 0 \text{ or } x = \frac{\pi}{2} \therefore$ for $0 < x < \pi, x = \frac{\pi}{2}$

at $x = \frac{\pi}{2}, \frac{d^2y}{dx^2} = \sin \pi^+ < 0.$
 at $x = \frac{\pi}{2}, \frac{d^2y}{dx^2} = \sin \pi^- > 0.$
 $\therefore$ Change in concavity.
 $\therefore$ point of inflection at $x = \frac{\pi}{2}.$



$\therefore V = \pi \left(\frac{1}{4} e^{-4x} \right) \Big|_0^2.$

$= -\frac{\pi}{4} e^{-8} + \frac{\pi}{4} e^0$

$V = \pi \int_0^2 (e^{-2x})^2 dx.$

$= \pi \int_0^2 e^{-4x} dx.$

$\therefore V_0 = \frac{\pi}{4} [1 - e^{-8}] \text{ units}^3$

Q6. a) $n = \frac{1}{2}$

$$\int_0^1 s dt \div \frac{1}{12} [0 + 55 + 2(52 + 49 + 50 + 52 + 53)]$$

$$\div \frac{1}{12} (55 + 512) \div \underline{47.25 \text{ km}}$$

b) Real roots $\Delta = b^2 - 4ac \geq 0$, $m^2 - 4 \times 2 \times 18 \geq 0$.

$$\therefore (m-12)(m+12) \geq 0 \quad \therefore \underline{m \leq -12 \text{ or } m \geq 12}$$

c) (i) $\alpha + \beta = \frac{3}{2}$ (ii) $\angle \beta = \frac{-5}{2}$

(iii) $(\alpha+1)(\beta+1) = \alpha\beta + (\alpha+\beta) + 1 = \frac{-5}{2} + \frac{3}{2} + 1 = 0$

d) $A_1 = PR - M$ $\therefore 0 = 20000(1.015)^{60} - m \left(\frac{1.015^{60} - 1}{0.015} \right)$

$$A_2 = PR^2 - M(1+R)$$

$$A_3 = PR^3 - M(1+R+R^2)$$

$$\therefore m = \frac{20000(1.015)^{60} \times 0.015}{1.015^{60} - 1}$$

$$\therefore A_n = PR^n - m \left(\frac{R^n - 1}{R - 1} \right) \quad \therefore \underline{m = \$571.43}$$

Q7. a) $r_1 = \frac{3}{4} \div \frac{9}{3} = \frac{2}{3}$, $r_2 = \frac{1}{2} \div \frac{3}{4} = \frac{2}{3} \therefore r$ is common.

$|r| < \frac{2}{3} \therefore$ Limiting sum of a GP exists.

$$S_{\infty} = \frac{a}{1-r} = \frac{9}{8} \div \frac{1}{3} = \underline{\frac{27}{8} \approx 3\frac{3}{8}}$$

b) (i) $U_5 = 13$, $U_6 = 15$, $U_7 = 17$, $U_{20} = 43$.

(ii) $U_6 - U_5 = 2$, $U_7 - U_6 = 2 \therefore$ common difference $\therefore$ A.P.

(iii) $\sum_{n=5}^{20} (2n+3) = \frac{n}{2}(a+l) = \frac{16}{2}(13+43) = \underline{448}$

c) $U_1 = 2+n$ $\frac{5+n}{2+n} = \frac{9+n}{5+n} \therefore 25+10n+n^2 = 18+11n+n^2$

$$U_2 = 5+n$$

$$U_3 = 9+n \quad (5+n)^2 = (2+n)(9+n) \therefore \underline{n=7}$$

Q8. a) $V = 3t(4-t) = 12t - 3t^2$ $a = 12 - 6t$

(i) at $t=1$ $a = 6 \text{ m. sec}^{-2}$

(ii) $x = \int v dt = 6t^2 - t^3 + c \therefore \underline{x = 6t^2 - t^3}$

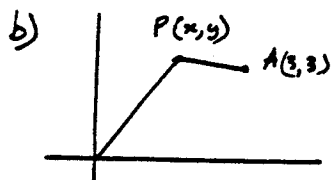
(iii) $V = 0$ $t = 4$ $x = 6(4)^2 - 4^3 = 2(16) = \underline{32 \text{ m}}$

(iv) $x = 0$ $t = 6$ $v = 18(-2) = \underline{-36 \text{ m. sec}^{-1}}$

(v) Max V when $a = 0 \therefore 12 - 6t = 0$ $t = 2 \text{ Sec}$

(vi) when $t = 5$ $x = 6(5^2) - 5^3 = \underline{25 \text{ m}}$

$$\therefore \text{Distance} = 32 + (32 - 25) = \underline{39 \text{ m}}$$



$$PO = 2PA$$

$$PO^2 = 4(PA)^2$$

$$x^2 + y^2 = 4(x-3)^2 + 4(y-3)^2$$

$$x^2 + y^2 = 4x^2 - 24x + 36 - 4y^2 + 24y - 36$$

$$3x^2 - 24x + 36 + 3y^2 - 24y + 36 = 0$$

$$x^2 - 8x + y^2 - 8y = -24$$

$$(x-4)^2 + (y-4)^2 = 32 - 24 = 8$$

Q9. a) (i) $\cos \theta = \frac{4^2 + 4^2 - 2^2}{2 \times 4 \times 4} = \frac{28}{32} = \frac{7}{8}$

$$\therefore \theta = 28^\circ 57'$$

(ii) $BC = 1 \text{ m}$

(iii) $4x^2 = 4^2 - 1$
 $4x^2 = 15$
 $\therefore x = \frac{\sqrt{15}}{2}$

(iv) $\frac{h}{8} = \frac{\sqrt{15}}{4}$
 $\therefore h = 2\sqrt{15}$

(v) $A = \frac{3+7}{2} 2\sqrt{15} - \frac{1}{2} \left(\frac{\sqrt{15}}{2} \right) - \frac{2+3}{2} \left(\frac{\sqrt{15}}{2} \right)$

$$\therefore \text{Area} = \frac{17}{2} \sqrt{15} \text{ m}^2$$

b) amp = 3 (ii) Period = 3π

(iii) $y = -3 \sin\left(\frac{2}{3}x\right)$

Q10. (i) $A = \int_0^3 3x - x^2 dx = \left[\frac{3}{2}x^2 - \frac{1}{3}x^3 \right]_0^3 = \frac{27}{2} - \frac{27}{3} = \underline{\frac{9}{2} \text{ u}}$

(ii) $A = \int_0^2 (3x - x^2) - x dx = \int_0^2 2x - x^2 dx$
 $= \left[x^2 - \frac{1}{3}x^3 \right]_0^2 = 4 - \frac{1}{3}(8) = \underline{\frac{4}{3} \text{ units}^2}$

(iii) $\% = \frac{4}{3} \div \frac{9}{2} \times 100\% = 29.63\% \approx 30\%$

b) (i) $\ell = r\theta$ (ii) $A = \frac{1}{2}r^2\theta$

c) $P = 20 = x + x + 2x\theta + 3x\theta \therefore 2x + 5x\theta = 20$

(i) $A = \frac{\theta}{2}(3x)^2 - \frac{\theta}{2}(2x)^2 \therefore \theta = \frac{20-2x}{5x}$

$$= \frac{65x^2}{2}$$

$$A = \frac{5x^2}{2} \left(\frac{20-2x}{5x} \right)$$

$$= x(10-x)$$

$$\therefore A = 10x - x^2$$

(ii) $\frac{dA}{dx} = 10 - 2x = 0$

$$\therefore \text{MAX Area}$$

$$\therefore \underline{x = 5}$$

$$h = 50 - 25$$

$$\frac{d^2A}{dx^2} = -2 \therefore \text{cc down}$$

$$\text{Area is } 25 \text{ unit}^2$$

$$\therefore \text{max T.P.}$$